The Methane-Oxidizing Bacteria (Methanotrophs) 14

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Contents

Abstract

Aerobic methane-oxidizing bacteria (methanotrophs) have the unique ability to grow on methane as their sole source of carbon and energy. They are ubiquitous in the environment and play a major role in the removal of the greenhouse gas methane from the biosphere before it is released into the atmosphere. The ability to drive oxygen-dependent methane oxidation was once assumed to be an exceptional property of a very restricted set of microbes belonging to two classes

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of *Proteobacteria*: *Alphaproteobacteria* and *Gammaproteobacteria*. While *Proteobacteria* still form the foundation of the methanotrophic landscape in many ecosystems, the ability to oxidize methane has also been demonstrated in the microbial phyla *Verrucomicrobia* and *Candidatus Methylomirabilis oxyfera* (phylum NC10). Over the years various methanotrophs have also been isolated, including facultative methanotrophs, extremophile species, and anaerobes, thus expanding both the taxonomic diversity and physiological range of methanotrophy. In addition, a number of cross-species interactions that enable efficient methane utilization have been identified, changing the way we view mechanisms of methane utilization. Finally, a thorough revision of core metabolic pathways has been made, and whole-genome metabolic models have been constructed, which facilitate the metabolic engineering of methanotrophic bacteria and expand the potential for their biotechnological applications.

1 Introduction

Methane-oxidizing bacteria or methanotrophs are a group of bacteria that grow on methane as their sole source of carbon and energy. They are a subset of methylotrophic bacteria, which can grow on a number of different one-carbon compounds (Lidstrom 2006). They appear to be ubiquitous in the environment and can be isolated from many different environments, including anoxic zones. They play a major role in the consumption of the atmospheric greenhouse gas methane as well as capturing biologically or geothermally formed methane before it is released into the atmosphere. All described methanotrophic bacteria contain the enzyme methane monooxygenase which oxidizes methane to methanol (reviewed in Lawton and Rosenzweig 2016), which is subsequently oxidized via formaldehyde and formate to carbon dioxide. All methanotrophs use methane to generate reducing power for biosynthesis. Methanotrophs contain specialized metabolic pathways for the assimilation of carbon, which can be incorporated into biomass at the level of formaldehyde (*Gammaproteobacteria*), formate (*Alphaproteobacteria*), or CO_2 (*Alphaproteobacteria*, *Verrucomicrobia*, and NC10). Different metabolic arrangements impact the overall carbon conversion efficiency (summarized in Kalyuzhnaya 2016; Table 1).

There has been considerable interest over the years in the use of methanotrophs or methanotrophic enzymes for the production of single-cell protein (SSP) and polyhydroxybutyrate (PHB), biofuels, and chemicals (reviewed in Strong et al. 2016). Until recently it was thought that methanotrophs could only grow on one-carbon compounds, but now a methanotroph that grows on multicarbon compounds has been characterized (see below). Methanotrophs were first described in the literature over 100 years ago (reviewed in Hanson and Hanson 1996), but it was the pioneering studies of Roger Whittenbury and colleagues that resulted in the isolation and characterization of over 100 methanotrophs from different environments (Whittenbury et al. 1970a). Readers are referred to these original papers for an extensive collection of excellent light micrographs and electron microscopy photographs which detail the morphological features of a large number of different methanotrophs (Davies and Whittenbury 1970; Whittenbury et al. 1970a, b). The characterization of these methanotrophs provided a classification scheme and taxonomic framework that remained remarkably robust over the following 30 years or so. Bowman and colleagues provided a detailed taxonomic framework for obligate methane-oxidizing bacteria, based largely on the original taxonomy system of Whittenbury and colleagues (Bowman et al. 1993). Since then, many new species of methanotrophic bacteria have been isolated. The taxonomy of the aerobic methane-oxidizing bacteria has recently been summarized in a series of publications in Bergey's Manual of Systematics of Archaea and Bacteria (http://onlinelibrary.wiley.com/book/10.1002/9781118960608) and thus will be only briefly described below.

2 Taxonomy/Phylogeny

The phylogenetic relationship of methane-oxidizing bacteria is shown in Fig. 1. Methanotrophic bacteria can be divided into four main groups: *Gammaproteobacteria* (often referred as Type I or Type X, Fig. 2), *Alphaproteobacteria*

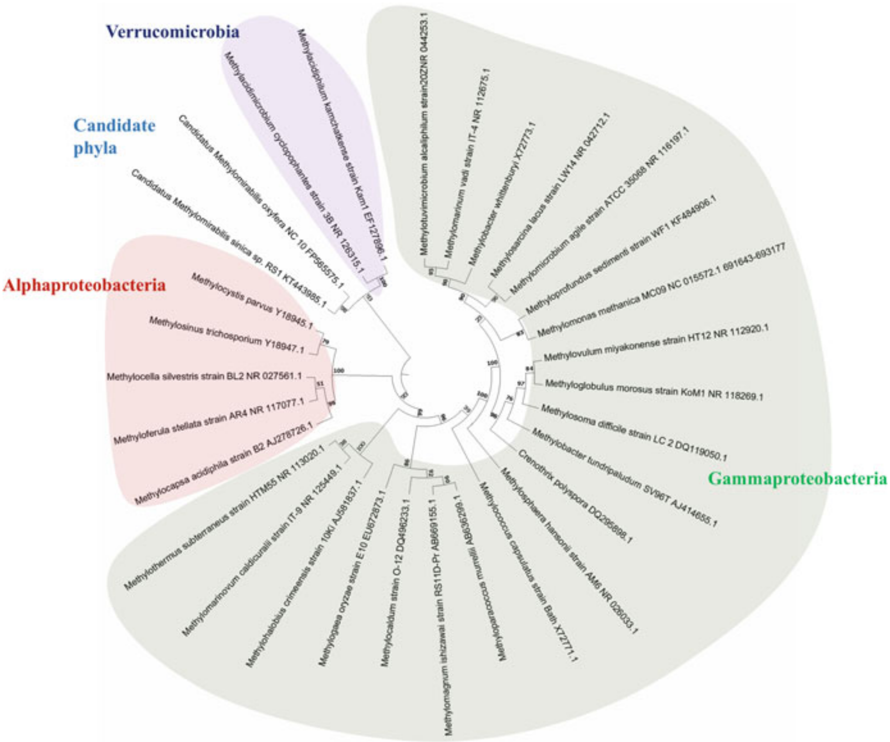


Fig. 1 16S ribosomal RNA gene phylogeny of methane-oxidizing bacteria

Fig. 2 Electron micrograph of a cross-section of a typical Type I methanotroph *Methylomonas methanica* showing characteristic bundles of intracytoplasmic membranes. (Photograph courtesy of H. Dalton) × 65,000 magnification

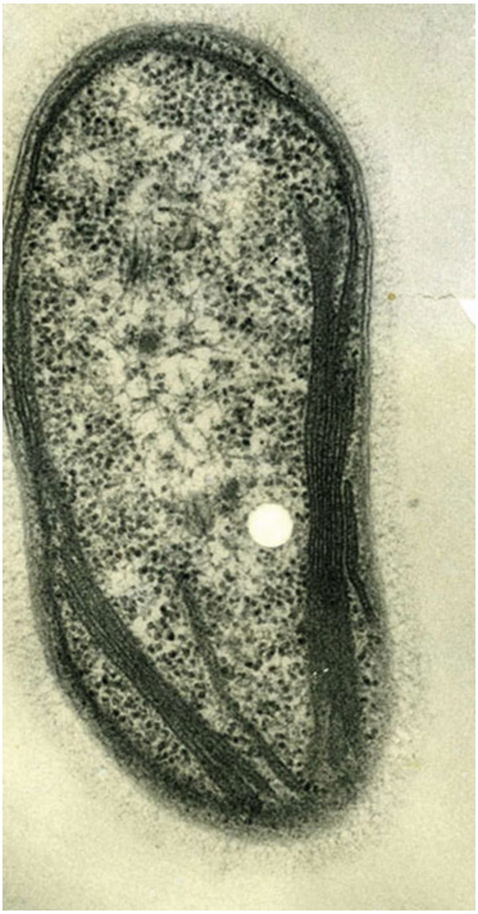
(previously known as Type II, Fig. 3), *Verrucomicrobia*, and members of NC10 phylum. Many groups of *Proteobacteria* and *Verrucomicrobia* phyla are represented by axenic cultures. However, many species of methanotrophic bacteria are resistant to cultivation in lab settings. The filamentous bacteria, *Crenothrix polyspora* and *Clonothrix fusca,* are examples of fastidious methanotrophs that cannot be obtained in pure culture (Vigliotta et al. 2007).

The genetic signature of these filamentous bacteria suggests that they are somewhat distant relatives of more conventional members of *Methylococcaceae.* Furthermore, the NC10 phylum is currently represented by a highly enriched consortia of methane-utilizing *Candidatus Methylomirabilis oxyfera* and satellite heterotrophs (Ettwig et al. 2009) and microcolonies of *Candidatus Methylomirabilis sinica* (He et al. 2016).

3 Morphology and Cell Biology

It is well documented that methane oxidation in most *Proteobacteria* and some *Verrucomicrobia* is associated with the formation of dedicated, subcellular compartments (intracytoplasmic membranes, ICMs, Figs. 2 and 3). Typically, gammaproteobacterial methanotrophs contain bundles of intracytoplasmic membranes (Fig. 2),

Fig. 3 Electron micrograph of a cross-section of a typical Type II methanotroph *Methylosinus trichosporium* showing characteristic intracytoplasmic membranes arranged around the periphery of the cell. (Photograph courtesy of H. Dalton) × 75,000 magnification



while alphaproteobacterial methanotrophs generally contain intracytoplasmic membranes arranged around the periphery of the cell (Fig. 3). ICMs can readily be visualized by electron microscopy in cells grown on methane in the presence of copper ions. Starvation, methanol growth, or copper-limiting conditions can lead to ICM degradation or reduced formation (Tavormina et al. 2016). The ICMs house the particulate methane monooxygenase (pMMO), the electron transport complexes, and ATP synthase. Some studies suggest that ICMs are formed via invagination of inner membrane, although the mechanisms of biogenesis and regulation of synthesis warrant further study.

S-layers are among other cell structural elements which have been described for some methanotrophic species (e.g., *Methylomicrobium, Methylotuvimicrobium* and *Methylococcus*) (Khmelenina et al. 2013b). While structural elements of S-layers are well documented, the genes and the enzymes involved in their synthesis and their functional roles remain to be discovered. Many gammaproteobacterial methanotrophs, particularly some *Methylobacter* species, form resting stages as *Azotobacter*-like cysts which develop in stationary-phase cultures. These cyst-like resting stages appear to confer advantages to methanotrophs, including increased survival upon desiccation. Although this is not a unifying feature of all genera of *Gammaproteobacteria* methanotrophs (perhaps the ability to form resting stages has been lost in some strains because of long-term cultivation in the laboratory), it is quite distinctive in some stationary-phase cultures of methanotrophs (see Bowman 2006). Alphaproteobacterial methanotrophs, such as *Methylosinus* and *Methylocystis*, form exospores or lipoidal cysts. The exospores, which bud off from vegetative cells during stationary phase, are resistant to heat (85 °C for 10 min) and can survive periods of drying for 18 months (Bowman 2006). Resting forms, S-layers, and carbohydrate-rich capsules have been discussed as possible targets for biotechnological applications, as possible sources of alginate, nanomaterial, or feedstock (Strong et al. 2016).

4 Physiology and Biochemistry

The main characteristics of aerobic methanotrophs are summarized in Table 1. All methanotrophic bacteria use oxygen for methane activation (Trotsenko and Murrell 2008; Sirajuddin and Rosenzweig 2015) and oxidize methane to methanol via the enzyme methane monooxygenase (MMO) (Anthony 1982; Dalton 2005; Smith and Murrell 2010). Methanotrophs can possess two structurally and biochemically distinct forms of MMO, particulate (pMMO) and soluble methane monooxygenase (sMMO). pMMO is a copper-containing enzyme that is associated with the ICMs. sMMO is a cytoplasmic non-heme iron enzyme complex. Some of the best characterized methanotrophs, *Methylococcus capsulatus* (Bath) and *Methylosinus trichosporium* OB3b, can produce either form of MMO (reviewed in Murrell et al. 2000; Hakemian and Rosenzweig 2007; Ross and Rosenzweig 2017). The single factor that governs expression of the two types of MMO in these organisms is the concentration of available copper. At a high copper-to-biomass ratio, pMMO is produced, whereas sMMO is expressed only when the copper-to-biomass ratio during growth is low. Many methanotrophs such as the *Methylomonas methanica* S-1, *Methylomicrobium alcaliphilum* 20Z (now *Methylotuvimicrobium alcaliphilum* (Orata et al. 2019)), and *Methylomicrobium album* BG8 possess only pMMO. pMMO enzymes, encoded by *pmo* genes (*pmoCAB*), are related to ammonia monooxygenases (AMO, encoded by *amoCAB*); however, the phylogeny of pMMO and AMO enzymes, distributed among diverse bacterial species, is complicated (Tavormina et al. 2011; Coleman et al. 2012; Osborne and Haritos 2018). In addition to *pmoCBA* or *amoCBA* genes with ascribed functions, related sequences are sometimes found in methanotroph genomes in addition to *pmoCAB* gene clusters; however, their function remains undetermined (Tavormina et al. 2011). Furthermore, *pmo*-like gene clusters (*hmoCAB*) in *Actinobacteria* have been linked to oxidation of C_n-alkanes, such as ethane, propane, or butane (Coleman et al. 2012; Wang et al. 2017). Overall, the genetic diversity and distribution of *pmo*-like genes within bacteria is significant (Osborne and Haritos 2018), and the precise roles of

Table 1 General summary of existent methanotrophic cultures

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
Gammaproteobacteria								
Methylococcaceae								
Clonothrix	*C. fusca*	0.01 (when grown with formaldehyde)	pMMO, Mxa	5	(10)	Forms attached or free filaments, surrounded by a yellowish brown distinctive sheath that may be encrusted	**16S rRNA partial sequence** [DQ984190.1]	(Vigliotta et al. 2007)
Crenothrix	*C. polyspora*	NA	BS, dPPP, FDH, H_4F, H_4MTP, pMMO, RuMP, sMMO, Xox	NA	(5)	Can oxidize methane under oxygen-deficient conditions; can reduce nitrate to nitrous oxide	**Not available as a pure culture** [NZ_FUKI00000000.1]	(Cohn 1870; Stoecker et al. 2006; Oswald et al. 2017)
Methylobacter	*M. luteus*	0.12	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP, TCA, Xox	5–9 (6.8)	(37)	Produces yellow water-soluble pigment and bacteriocin-like compounds	**VKM-53B** ^{T}* [NZ_ATYJ00000000.1]	(Starostina et al. 1998; Hamilton et al. 2015)
	M. marinus	0.15	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, pXmo RuMP, TCA, Xox	5.5–9 (7)	15–40 (35)	Requires NaCl concs. of 0.1–5%; can form desiccation-resistant cysts	**A45**^{T}* [NZ_ARVS00000000.1]	(Bowman et al. 1993; Flynn et al. 2016)

(*continued*)

Table 1 (continued)

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
	M. psychrophilus	NA	pMMO	6–7.6 (6.8–7.2)	4–20 (10)	Psychrophilic strain	**Z-0021^{T}* 16S rRNA sequence** [NR_025016.1]	(Omel'chenko et al. 1996)
	M. tundripaludum	0.02	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC RuMP	5.5–8 (6.8)	5–30 (23)	Poor to no growth on methanol	**SV96^{T}*** [NZ_AEGW00000000.2]	(Wartiainen et al. 2006b)
	M. whittenburyi	0.1	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP, TCA, Xox	5–8 (6.8)	5–40 (25)	Does not grow on methanol	**UCM-B-3033^{T}*** [NZ_JQNS00000000.1]	(Whittenbury et al. 1970a; Hamilton et al. 2015)
	Methylobacter sp. BBA5.1	0.06	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, pXmo, RuMP, TCA, Xox	NA (7.2)	5–35 (30)	Grows at NaCl concs. of 0.1–5%	**BBA5.1** [NZ_JQKS01000001.1]	(Smith et al. 1997; Flynn et al. 2016)
Methylocaldum	*M. szegediense*	0.03	BS, CBB, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP	NA (6.8)	30–61 (55)	Moderately thermophilic; produces melanin-like pigments and can excrete 4-hydroxyphenyl acetic acid; can produce PHB	**O-12** [NZ_ATXX00000000.1]	Eshinimaev et al. 2004; Medvedkova et al. 2007)

	M. marinum	0.02	BS, CBB, dPP, FDH, H_4F, H_4MTP pMMO, RuMP, SC-EMP, sMMO	6–8 (7)	20–47 (36)	Requires NaCl for growth, optimal growth at 2% NaCl (w/v); growth with methanol at 0.01–2% (v/v)	**S8**^{T}***** [AP017928.1]	(Takeuchi et al. 2014)
Methylococcus	*M. capsulatus* Bath	0.3–0.4	BS, CBB, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP, sMMO, TCA	6–8 (7)	30–55 (42)	Well-established strain for single-cell protein production	**Bath**^{T}***** [NC_002977.6]	(Whittenbury et al. 1970a; Ward et al. 2004)
	M. capsulatus Texas	0.1	BS, CBB, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP, sMMO, TCA	6–8 (7)	30–50 (37)	Thermotolerant; grows well on methanol	**Texas** [NZ_AMCE00000000.1]	(Foster and Davis 1966; Kleiveland et al. 2012)
Methylogaea	*M. oryzae*	0.14	BS, CBB, dPPP, EDD, EMP, FDH, H_4MTP, Mxa, pMMO, pSC, RuMP	5–8 (6.6)	20–37 (30)	Sensitive to NaCl concs. above 0.5% (w/v); grows on methanol at concs. of 0.1–1% (w/v); can produce PHB	**E10**^{T}***** [BBDL00000000.1]	(Geymonat et al. 2011)

(*continued*)

Table 1 (continued)

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
Methyloglobulus	*M. morosus* KoM1	N/A	BS, dPPP, EDD, EMP, FDH, H_4MTP, pMMO, pSC, RuMP, TCA	5–8.5 (6–8)	4–30 (20)	No growth at NaCl concs. above 0.5% (w/v); can fix nitrogen; grows best at reduced oxygen concs. (0.05–0.1 bar)	**KoM1^{T}*** [NZ_AYLO00000000.1]	(Poehlein et al. 2013; Deutzmann et al. 2014)
Methylomagnum	*M. ishizawai*		BS, CBB, FDH, H_4F, H_4MTP, Mxa, pMMO, PPP, RuMP, sMMO, TCA, Xox	5.5–9 (6.8–7.4)	20–37 (32)	Good growth with 0–0.2% (w/v) NaCl and 0–0.05% methanol (v/v)	**RS11D-Pr^{T}*** [FXAM00000000.1]	(Khalifa et al. 2015; Frindte et al. 2017)
Methylomarinum	*M. vadi* IT-4	0.33	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, PPP, RuMP, TCA, Xox	4.5–7.0 (6.4)	20–44 (37)	Requires NaCl for growth, optimal salinity is of 2–3% NaCl (w/v); produces PHB	**IT-4^{T}*** [NZ_JPON00000000.1]	(Hirayama et al. 2013; Flynn et al. 2016)
Methylomicrobium	*M. agile* A30	0.06	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP	6–9 (7)	10–37 (27)		**A30^{T}*** [NZ_JPOJ00000000.1]	(Whittenbury et al. 1970a; Foster and Davis 1966)
	M. album BG8	0.11	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP, TCA, Xox	6–9 (7)	10–37 (27)		**BG8^{T}*** [NZ_AFJF00000000.2]	(Whittenbury et al. 1970a; Kits et al. 2013)

	M. lacus LW14	0.05	BS, dPPP, FDH, H_4F, H_4MTP, pMMO, pSC, RuMP	NA (6.8)	NA (6.8)		**LW14^T*** [NZ_AZUN00000000.1]	(Kalyuzhnaya et al. 2015a)
Methylotuvimicrobium	*M. alcaliphilum* 20Z	0.12	BS, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, PPP pSC, RuMP, TCA	6.5–10 (9)	8–37 (30)	Optimal growth at NaCl concs. of 3% (w/v); accumulates ectoine and sucrose to promote osmotolerance resistant to desiccation	**20Z^T*** [NC_016112.1]	(Khmelenina et al. 1997; Kalyuzhnaya et al. 2008; Orata et al., 2019; Vuilleumier et al. 2012)
	M. buryatense 5G	0.23	BS, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, PPP, RuMP, sMMO, TCA	6–11 (9)	4–45 (28–30)	Optimal growth at 0.5–0.75% NaCl (w/v); resistant to desiccation	**5B^T*** [NZ_AOTL00000000.1]	(Kalyuzhnaya et al. 1999; Khmelenina et al. 2013)
Methylomonas	*M. methanica* MC09	NA	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, RuMP, sMMO, TCA	6.6–8	(30)	Strain was isolated from seawater, grows at 5% NaCl (w/v)	**MC09** [NC_015572.1]	(Dworkin and Foster 1956; Boden et al. 2011)
	Methylomonas LW13	0.16	BS, dPPP, EDD, EMP, FDH, H_4F, H_4MTP, Mxa, pMMO, RuMP, SC, sMMO	NA (6.8)	NA (30)	Does not require NaCl for growth	**LW13** [JNLB00000000.1]	(Auman et al. 2000; Kalyuzhnaya et al. 2015a)

(*continued*)

Table 1 (continued)

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
	M. paludis	NA	FDH, Mxa, pMMO, RuMP	3.8–7.3 (5.8–6.4)	8–30 (22)	Will not grow at NaCl concs. above 0.1% (w/v); optimal methanol concs. of 0.25% (v/v)	**MG30^{T}*** **16S partial rRNA sequence** [NR_108887.1]	(Danilova et al. 2013)
	M. lenta	NA	BS, dPPP FDH, pMMO	6.3–7.8 (7.0)	15–28 (20)	Slow-growing strain; sensitive to methanol concs. ≥0.5% (v/v)	**R-45377^{T}*** [NZ_LUUI00000000.1]	(Hoefman et al. 2014a)
Methyloparacoccus	*M. murrellii*	NA	Mxa, pMMO	5.8–9 (6.5)	20–37 (25–33)	Sensitive to addition of 1% NaCl (w/v)	**LMG 27482^{T}*** **16S partial rRNA sequence** [AB636299.1]	(Hoefman et al. 2014b)
Methyloprofundus	*M. sedimenti*	0.08	BS, dPPP, FDH, H_4F, H_4MTP, pMMO, pSC, RuMP	6–8 (7.0)	4–26 (23)	Requires NaCl for growth, optimal concs. being 2% (w/v)	**WF1^{T}*** [NZ_LPUF00000000.1]	(Tavormina et al. 2015)
Methylosarcina	*M. fibrata* AML-C10	0.08	BS, dPPP, FDH, H_4F, H_4MTP, pMMO, pSC, RuMP	5–9 (5)	10–37 (30)	Can grow at NaCl concs. of up to 1% (w/v); grows poorly with methanol	**AML-C10^{T}*** [NZ_ARCU00000000.1]	(Wise et al. 2001)

Methylosoma	*M. difficile* LC 2	0.01	pMMO	5–9 (6–8)	16–30 (25)	NaCl concs. above 0.5% (w/v) inhibit growth; grows well in dilute NMS medium (1/10 conc. of potassium nitrate) and prefers low oxygen tensions; can fix nitrogen	**LC 2^{T}*** **16S partial rRNA sequence** [DQ119050]	(Rahalkar et al. 2007)
Methylosphaera	*M. hansonii*		RuMP		0–21 (10)	Requires artificial seawater salts for growth (at concs.70–100%); can grow on methanol and can fix nitrogen	**AM6^{T}*** **16S partial rRNA sequence** [NR_026033.1]	(Bowman et al. 1997)
Methylovulum	*M. miyakonense* HT12	0.01	EDD, EMP, dPPP, FDH, H_4F, H_4MTP, Mxa, pMMO, pSC, sMMO, RuMP, TCA, Xox	6–7.5	5–34 (24)	High sensitivity to NaCl, growth inhibited at concs. above 0.2% (w/v)	**HT12^{T}*** [NZ_AQZU00000000.1]	(Iguchi et al. 2011; Hamilton et al. 2015)
Methylothermaceae								
Methylohalobius	*M. crimeensis*	0.03	BS, CBB, EMP, FDH, H_4MTP, H_4F, Mxa, pMMO, pSC, PPP, RuMP, TCA	6.5–7.5	15–42 (30)	Grows optimally at NaCl concs. of 5.8–8.7% (w/v); evidence of PHB production	**10Ki^{T}*** [NZ_ATXB00000000.1]	(Heyer et al. 2005; Sharp et al. 2015)

(*continued*)

Table 1 (continued)

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
Methylomarinovum	*M. caldicuralii*	0.29–0.33	pMMO, RuMP	5–7 (6.4)	30–55 (50)	Optimal growth at NaCl concs. of 3% (w/v)	**IT-9^{T}*** **16S rRNA partial sequence** [NR_125449.1]	(Hirayama et al. 2014)
Methylothermus	*M. subterraneus*	0.3	pMMO, RuMP	5.2–7.5 (6.3)	37–65 (60)	Optimal growth at NaCl concs. of 0–0.3% (w/v)	**HTM55 ^{T}*** **16S rRNA partial sequence** [NR_113020.1]	(Hirayama et al. 2011)
Alphaproteobacteria								
Methylocystaceae								
Methylocystis	*M. rosea* SV97	NA	FDH, H_4F, H_4MTP, pMMO, SC-EMP	5.5–9.0	5–37 (27)	Does not require NaCl for growth; produces PHB	**SV97^{T}*** [NZ_ARCT00000000.1]	(Wartiainen et al. 2006b)
	Methylocystis ATCC 49242 strain Rockwell	0.16	FDH, H_4F, H_4MTP, Mxa, pMMO, SC-EMP	NA (6.8)	NA (30)	Highly tolerant of ammonium, capable of nitrogen fixation	**ATCC 49242** [AEVM00000000.2]	(Nyerges et al. 2010; Stein et al. 2011)
	M. parvus OBBP	0.12	FDH, H_4F, H_4MTP, Mxa pMMO, SC-EMP	6–8 (6.8)	5–37 (30)	Can produce PHB	**OBBP^{T}*** [NZ_AJTV00000000.1]	(Whittenbury et al. 1970a; del Cerro et al. 2012)
	Methylocystis sp. SC2	NA	FDH, H_4F, H_4MTP, Mxa, pMMO, SC-EMP	NA	NA	Low-affinity pMMO	**SC2** [HE956757.1]	(Dam et al. 2012)

	Methylocystis sp. SB2		FDH, H_4F, H_4MTP, Mxa, pMMO, PPP, SC-EMP, TCA	6–9 (6.8)	10–30 (30)	Facultative methanotroph with EMP pathway	**SB2** [NZ_AYNA00000000]	(Vorobev et al. 2014)
Methylosinus	*M. trichosporium OB3b*	0.12–0.14	FDH, H_4F, H_4MTP, Mxa, pMMO, SC-EMP, sMMO	6–8 (6.8)	5–37 (30)	Well-established catalysis for PHB production, epoxidation; can degrade chlorinated aliphatic hydrocarbons; can produce methanobactins	**OB3b ^{T}*** [NZ_ADVE00000000.2]	(Whittenbury et al. 1970b; Oldenhuis et al. 1989; Stein et al. 2010)
	Methylosinus sp. LW4	0.12–0.15	FDH, H_4F, H_4MTP, pMMO, SC-EMP, sMMO	6–8 (6.8)	5–37 (30)	Strain shows robust growth; can produce methanobactins	**LW4** [ARAB00000000.1]	(Auman et al. 2001; Kenney et al. 2016)
Beijerinckiaceae								
Methylocella	*M. silvestris*	0.014	dPPP, FDH, H_4F, H_4MTP, Mxa, SC-GS, sMMO, TCA	5–7.0 (5.5)	4–30 (25)	Facultative methanotroph; growth is inhibited by NaCl concs. above 0.8% (w/v)	**BL2 ^{T}*** [NC_011666.1]	(Dunfield et al. 2003; Chen et al. 2010)
	M. tundrae	NA	FDH, Mxa, PPP, SC-GS, sMMO, TCA	4.2–7.5 (5.5)	5–30 (15)	Growth is inhibited by NaCl concs. above 0.8% (w/v); best growth at methanol concs. of 0.5–1% (v/v)	**T4 ^{T}*** **16S rRNA partial sequence** [NR_025596.1]	(Dedysh et al. 2004)

(*continued*)

Table 1 (continued)

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
Methylocapsa	*M. acidiphila B2*	0.03	BS, FDH, H_4F, H_4MTP, Mxa, pMMO, PPP, SC-GS, TCA	4.2–7.2 (5.5)	10–30 (24)	NaCl inhibits growth at a concs. of 0.5% (w/v); methanol supports growth at concs. below 0.05% (v/v); capable of PHB production	**B2**^{T*} [NZ_ATYA00000000.1]	(Dedysh et al. 2002)
	M. aurea KYG	0.018	BS, FDH, H_4F, H_4MTP, Mxa, pMMO, SC-GS, TCA	5.2–7.2 (6.2)	2–33 (30)	Growth is inhibited with 0.3% (w/v) NaCl; methanol supported growth at concs. below 0.1% (v/v); capable of PHB production	**KYG**^{T*} [JQKO00000000.1]	(Dunfield et al. 2010)
Methyloferula	*M. stellata* AR4	0.005 (CH_4)/ 0.015 (CH_3OH)	BS, CBB, FDH, H_4MTP, Mxa, SC-GS, sMMO, TCA, Xox	3.5–7.2 (5.2)	4–33 (23)	Reduced growth at NaCl concs. above 0.7% (w/v). Optimum methanol concs. between 0.1 and 1% (v/v)	**AR4**^{T*} [NZ_ARWA00000000.1]	(Vorobev et al. 2011; Dedysh et al. 2015)

Candidate phylum NC10								
Candidatus Methylomirabilis								
Methylomirabilis	*M. oxyfera*		CBB, FDH, H_4MTP, Mxa, pMMO	7–8	25–30	Intracellular oxygen production; requires nitrite for growth	**No pure strains** [FP565575.1]	(Ettwig et al. 2010; Rasigraf et al. 2014)
Verrucomicrobia								
Methylacidiphilaceae								
Methylacidiphilum	*M. fumariolicum* SolV	0.07–0.08	BS, CBB, EMP, FDH, pMMO, PPP, TCA, Xox	0.8–5.8 (2)	40–65 (55)	Grows at very low pH; capable of nitrogen fixation	**SolV** [PRJEA85607]	(Pol et al. 2007; Khadem et al. 2011; Anvar et al. 2014)
	M. infernorum V4	0.04	BS, CBB, FDH, pMMO, Xox	1–6 (2–2.5)	60	Capable of nitrogen fixation	**V4** [NC_010794.1]	(Dunfield et al. 2007; Hou et al. 2008)
	M. kamchatkense Kam1	0.01	CBB, EMP, FDH, pMMO, PPP, TCA Xox	2–5 (3.5)	40–60 (55)	Grows in low concs. of methanol (up to 0.0001% w/v); can fixate nitrogen	**Kam1** [NZ_JQNX00000000.1]	(Islam et al. 2008; Erikstad and Birkeland 2015)
Unclassified								
Methylacidimicrobium	*M. cyclopophantes*	0.04	CBB, pMMO, Xox	0.6–6 (1.5–3)	20–49 (44)	ICM system that consists of membrane stacks orthogonal to the cell wall	**3B** **16S rRNA partial sequence** [NR_126315.1]	(Van Teeseling et al. 2014)
	M. tartarophylax	0.04	CBB, pMMO, Xox	0.5–5 (1–3)	20–43 (38)	Growth is inhibited by oxygen; no ICM system was observed	**$4AC^T$*** **16S rRNA partial sequence** [NR_126314.1]	

(*continued*)

Table 1 (continued)

Name	Species	Growth parameters					Culture collection ID [Genome Accession Number (NCBI)]	References
		Growth rate (h^{-1})	Metabolic pathways	pH (opt.)	T °C (opt.)	Others		
	M. fagopyrum	0.01	CBB, pMMO, Xox	0.6–6 (1.5–3)	20–39 (35)	No ICM system was observed	**3C^{T*}** **16S rRNA partial sequence** [NR_126313.1]	

^{T*}, type strain; BS, *Bifidobacterium* shunt; concs., concentrations; dPPP, dissimilatory pentose phosphate cycle; EDD, Entner-Doudoroff pathway; EMP, Embden-Meyerhof-Parnas pathway; FDH, formate dehydrogenases; H_4MTP, methanopterin-linked C1 transfer; H_4FP, folate-linked C1 transfer; ICM, intracytoplasmic membranes; Mxa, PQQ-linked methanol dehydrogenases; NA, data not available; PHB, polyhydroxybutyrate; NMS, nitrate mineral salts medium; pMMO, particulate methane monooxygenase; pSC, partial serine cycle; pXmo, methane/ammonia monooxygenase-related proteins of unknown function; RuMP, assimilatory ribulose monophosphate pathway; sMMO, soluble methane monooxygenase; TCA, tricarboxylic acid cycle; Xox, PQQ-linked methanol and formaldehyde dehydrogenases some distant homologs await experimental validation. The sMMO genes seem to be less widespread among the methanotrophs than pMMO genes. However, some methanotrophs of the family *Beijerinckiaceae* only encode sMMO (Dedysh and Dunfield 2014). Moreover, some of these organisms encode additional soluble alkane monooxygenases, with different substrate specificities (propane-specific rather than methane-specific), and these are related to both sMMO and other soluble di-iron center monooxygenases from versatile *Proteobacteria* and *Actinobacteria* (Crombie and Murrell 2014).

The product of methane oxidation is methanol, which is further oxidized to formaldehyde (Anthony 1982). One enzyme that carries out this function, a periplasmic pyrroloquinoline quinone (PQQ)-dependent, calcium-containing methanol dehydrogenase (MDH), was identified over a half century ago (Anthony and Zatman 1964, 1965, 1967a, b) and has now been studied in detail (Anthony 2004). The enzyme consists of two subunits, large and small, MxaF and MxaI, and is functionally linked to a dedicated cytochrome MxaG (Williams et al. 2006). Most known methanotrophs and many non-methanotrophic methylotrophs contain this enzyme and the necessary accessory functions (Chistoserdova and Lidstrom 2013). Rather surprisingly, a novel MDH has been recognized recently, whose single subunit, XoxF, is homologous to MxaF (Chistoserdova 2016). It requires lanthanides, which are rare earth elements, in its active center instead of calcium (Pol et al. 2014; Keltjens et al. 2014). Genetic investigations in model organisms indicated that rare earth elements are not only important for the activity of XoxF enzymes, but that they are also involved in the regulation of the alternative MDH enzymes, by decreasing expression of the *mxa* gene cluster and by increasing expression of *xox* genes (Vu et al. 2016; Chu and Lidstrom 2016; Gu et al. 2016). The formaldehyde oxidation step in the methane oxidation pathway can be catalyzed by several enzymes, including the tetrahydromethanopterin- (H_4MPT-) or tetrahydrofolate-linked and formaldehyde dehydrogenases (possibly also XoxF enzyme). The organisms operating the ribulose monophosphate (RuMP) pathway for formaldehyde assimilation can also oxidize formaldehyde via the oxidative branch of this cycle (Anthony 1982).

The pathways for assimilation of methane carbon have recently undergone a thorough revision through matching recent omics data to established knowledge, in combination with enzyme and metabolite characterization. The gammaproteobacterial methanotrophs typically use the RuMP pathway as their primary route of incorporation of carbon into cellular material. Thermodynamically, the RuMP pathway is the most efficient pathway for methane carbon assimilation. Different variants of the pathway have been described, and it has been shown that the pathway can also contribute to fermentative metabolism (Kalyuzhnaya et al. 2013). A pathway for glyoxylate regeneration by the serine cycle methylotrophs which do not possess a glyoxylate shunt (among them the alphaproteobacterial methanotrophs) remained a mystery for half a century (Anthony 2011) and has been solved only recently as the ethylmalonyl-CoA cycle (EMC; Erb et al. 2007). The discovery of this new pathway also prompted a reevaluation of the amount of carbon assimilated in the form of CO_2 through the multiple carboxylation reactions that are key to this metabolism (Erb et al. 2007; Chistoserdova et al. 2009; Yang et al. 2013). Interestingly, genomic evidence indicated that the gammaproteobacterial methanotrophs also encode the serine cycle, but not the EMC or glyoxylate shunt (Chistoserdova et al. 2005; Kalyuzhnaya et al. 2015b); the role of this variant serine cycle in these methanotrophs and its significance for methylotrophy metabolism remain to be determined. The carbon assimilation through the Calvin Benson Bassham (CBB) cycle has been now demonstrated in some methanotrophs, such as Verrucomicrobia and Candidate Phylum NC10 (Khadem et al. 2011; Rasigraf et al. 2014).

Thus far, the majority of methanotrophic cultures isolated in pure culture have been described as obligate, i.e., they can use only methane and methanol as sources of carbon and energy. However, the number of facultative methanotrophs (i.e., those able to use multicarbon compounds) has increased over time. Many newly described species fall into groups canonically associated with obligate methanotrophy (Semrau et al. 2008). The first fully authenticated facultative methanotroph, *Methylocella silvestris*, which can grow on either methane or multicarbon compounds, is found among *Alphaproteobacteria* (Dedysh et al. 2005; Im et al. 2010; Theisen et al. 2005). This methanotroph has been shown to be able to consume short-chain alkanes as an alternative or supplement to methane (Crombie and Murrell 2014).

The ability to fix N_2 by Type II methanotrophs and *Mc. capsulatus* (Bath) has been known for some time (Romanovskaia et al. 1980). Genomic evidence resulting from screening for the gene *nif*H, which encodes the Fe protein of nitrogenase, indicates that a number of gammaproteobacterial methanotrophs such as *Methylomonas*, *Methylocaldum*, *Methyloprofundus*, and *Methylobacter* may also fix N_2 (Auman et al. 2001). Type I methanotrophs assimilate NH_4^+ mainly by reductive amination of pyruvate and/or α-ketoglutarate, whereas Type II methanotrophs use the glutamate cycle, i.e., glutamine synthetase (GS) and the glutamine-oxoglutarate amidotransferase (GOGAT) system (see Trotsenko and Murrell 2008).

5 Ecology/Habitats

Methanotrophic bacteria appear to be ubiquitous in the environment and can be isolated from many environments where methane is present. The physiological range of methanotrophy encompasses aerobic and anaerobic zones, acid and alkaline ecosystems, freshwater and moderately saline (up to 24%) environments, and highly oligotrophic and highly contaminated niches. Methane-utilizing bacteria typically flourish at the aerobic-anaerobic interface, but many groups have been detected in a variety of anoxic ecosystems. Various mechanisms have been proposed for growth of methanotrophic bacteria in anoxic environments; among the mechanisms most broadly discussed are intracellular O_2-production (Ettwig et al. 2010), close interactions with cyanobacteria (Milucka et al. 2015), and denitrification (Cao et al. 2019).

Freshwater and sediments harbor a variety of different methanotrophs. Gammaproteobacterial methanotrophs, including *Methylobacter*, *Methylomonas*, *Methylosarcina*, *Methylococcus*, and *Methylosoma*, are particularly predominant in these environments. For example, using three independent techniques, Costello et al.

(2002) have estimated that Type I methanotrophs are ten–hundredfold more abundant than Type II methanotrophs in Lake Washington sediment. In Lake Constance, Type I methanotrophs also predominate (Rahalkar et al. 2007). The study of methanotrophs in marine water columns and sediments is somewhat limited. Species from the *Methylomonas*, *Methylomicrobium*, and *Methyloprofundus* genera have been isolated; however, molecular methods suggest that there remains a wide distribution of yet-to-be-cultivated marine methanotrophs, mainly from *Methylococcaceae*. Upland and forest soils represent the major sink of atmospheric methane, and soils from these environments usually have high affinity for methane. The most frequently detected *pmoA* sequences from upland soils [upland soil cluster α (USCα) and upland soil cluster γ (USCγ)] belong to *Methylocystis*. It is thought that USCα and USCγ belong to uncultivated methanotrophs which can utilize atmospheric concentrations of methane (discussed in Ricke et al. 2005; Knief 2015; Pratscher et al. 2018). Baani and Liesack (2008) have shown that multiple copies of genes encoding pMMO from *Methylocystis* have different affinities for methane and this may explain why *Methylocystis* is ubiquitous in forest and upland soils and a number of other environments. New evidence indicates the existence of *Gammaproteobacteria* methanotrophs with high affinity for methane (Edwards et al. 2017). In some ecosystems, high-affinity methane oxidation has been linked to the activity of conventional methanotrophic communities, consistent with both gamma- and alphaproteobacterial traits (Cai et al. 2016).

Rice cultivation is a major source of atmospheric methane, and its contribution to the global methane budget may continue to grow due to the increasing demand for rice. Studies have shown that rice fields contain highly diverse methanotrophs, including *Methylomonas*, *Methylobacter*, *Methylomicrobium*, *Methylococcus*, *Methylocaldum*, *Methylocystis*, and *Methylosinus*. The distribution and abundance of methanotrophs in rice fields are affected by several factors, including oxygen availability and the growth period of the rice (Eller and Frenzel 2001). No obvious geographical distribution patterns have been found for methanotrophs present in rice fields around the world, although *Methylocaldum* spp. tend to be detected in tropical regions.

Methanotrophs are also found in landfills, e.g., *Methylobacter*, *Methylosarcina*, *Methylomicrobium*, *Methylocystis*, and *Methylosinus*, with species of *Methylobacter* and *Methylocystis* being the most dominant (Cébron et al. 2007; Su et al. 2014).

Methanotrophs have been found in many extreme environments, ranging from 0 °C to 70 °C, salinity from 0% to 20% (NaCl), and pH 1.5–12. Pure cultures of thermophilic, psychrophilic, acidophilic, alkaliphilic, and halophilic methanotrophs have been obtained (reviewed in Trotsenko and Khmelenina 2002; Dunfield 2009). Thermophilic methanotrophs include *Methylococcus capsulatus* Bath which was isolated from a hot spring (Whittenbury et al. 1970a), *Methylocaldum* species (Bodrossy et al. 1997, 1999), and *Methylothermus thermalis,* which was isolated from a Japanese hot spring (Tsubota et al. 2005). Molecular methods suggest that the diversity of thermophilic methanotrophs is even greater, and many species remain to be uncultivated (Eloe-Fadrosh et al. 2016). Three isolated psychrophilic methanotroph strains include *Methylobacter psychrophilus* (Omel'chenko et al. 1996), *Methylosphaera hansonii* which was isolated from an Antarctic lake (Bowman et al. 1997), and *Methylomonas* *scandinavica* which was isolated from ground water of Sweden (Kalyuzhnaya et al. 1999). These Type I methanotrophs grow between 5 °C and 15 °C and have a low DNA G + C content (43–54 mol%). Halophilic and alkaliphilic methanotrophs have also been isolated, which mainly belong to the genus *Methylomicrobium* (Trotsenko and Khmelenina 2002). However, culture-independent methods have demonstrated that soda lakes harbor diverse methanotrophs other than *Methylomicrobium*. The isolation and characterization of acidophilic/acid-tolerant methanotrophs started with soils taken from peatlands and acidic forests (Dedysh et al. 2002), which represent one of the major sources of atmospheric methane. Two genera were isolated and validated, *Methylocella* and *Methylocapsa* (Dedysh et al. 2002; Dunfield et al. 2003). Another study reported that *Methylocystis* species may also be acidophilic/acid-tolerant (Dedysh et al. 2007). Three thermophilic and extremely acidophilic *Verrucomicrobia* species growing below pH 2 have been isolated and characterized (Dunfield et al. 2007; Islam et al. 2008; Pol et al. 2007). Two other methanotrophic genera, *Methylacidiphilum* and *Methylacidimicrobium* (Table 1), contain unusual *pmoA* sequences and may have atypical metabolic pathways for carbon metabolism and assimilation (reviewed briefly in Semrau et al. 2008). Initially described as extremophile thermoacidophilic methanotrophs, some members of *Verrucomicrobia* are now found to be mesophilic (Dunfield et al. 2007; Islam et al. 2008; Pol et al. 2007; reviewed in Semrau et al. 2008).

Methanotrophs are also found as symbionts (reviewed in Dubilier et al. 2008). The first methanotroph-based symbiosis to be discovered was in deep-sea mussels at cold seeps in the Gulf of Mexico. Subsequently symbiotic methane-oxidizing bacteria have been observed in other deep-sea mussels, hydrothermal-vent snails, tubeworms, and marine sponges. Although mainly observed in marine environments enriched in methane, methanotroph symbionts have also been found in a species of *Sphagnum* moss in peat bogs in the Netherlands. All of the methanotroph symbionts found in the marine symbioses are phylogenetically related to gammaproteobacterial methanotrophs (cf Type I), whereas the majority of discovered plant symbionts are related to free-living alphaproteobacterial methanotrophs. Thus far, all symbiotic methane oxidizers have resisted all attempts at cultivation in the laboratory. The subject of methanotroph symbioses is a very interesting one for future study and has been described in detail in Petersen and Dubilier (2009).

Cultivation-independent methods (reviewed in McDonald et al. 2008; Eloe-Fadrosh et al. 2016; Chistoserdova and Kalyuzhnaya 2018) have revealed that there are many more methanotrophs present in the environment that await isolation and cultivation in the laboratory. Future challenges will be in isolating methanotrophs that grow on atmospheric concentrations of methane, novel uncultivated species, as well as facultative methanotrophs.

6 Cultivation

Methanotrophs can be readily isolated from most environments using mineral salts medium and high-purity methane (natural gas contains a small proportion of ethane which can poison methanotrophs due to its conversion to ethanol and ethanol by

MMO). The nitrate mineral salts (NMS) medium first described by Whittenbury et al. (1970a) contains magnesium sulfate, potassium nitrate, and calcium chloride, supplemented by a trace element solution. Iron in the form of NaFeEDTA and molybdate are often also added, especially for the enrichment of nitrogen-fixing methanotrophs (minus inorganic nitrogen sources). Ammonia salts can substitute for nitrate as an inorganic nitrogen source but are usually added in lower concentrations since ammonia is a competitive inhibitor of MMO. Copper ions can also be added to ensure good growth since pMMO requires copper. Liquid cultures can be cultivated in flasks or serum vials sealed with airtight seals. Methane can then be injected into the headspace to typically 20% (v/v). The addition of rare earth elements (RRE, lanthanides) might stimulate isolation and growth of methanotrophic bacteria lacking Ca-dependent MDH (Pol et al. 2007). Tungsten supplementation might help to grow microbes with tungsten-dependent formate dehydrogenase or reduce formate excretion by pure cultures (Akberdin et al. 2018b). The addition of balanced micronutrients (Cu, Fe, W, rare earth elements, or Ca) can further improve yields of methanotrophic cells in large-scale cultivation. A number of strategies and optimized protocols for methanotroph cultivation are now available (reviewed in Dedysh SN, Dunfield PF (2014); Vekeman et al. 2017).

Methanotrophs can be grown on agar plates contained within airtight boxes (e.g., Tupperware) or anaerobic jars with a headspace of roughly 20% methane and 80% air. Methanotrophs can often be grown on methanol (high purity) although this is often toxic to some methanotrophs even at low concentrations (0.1%). Some methanotrophs can be "trained" to grow on methanol by building up methanol tolerance. Methanotrophs such as *Methylococcus capsulatus* Bath and *Methylosinus trichosporium* strain OB3b (the two best characterized methanotrophs) can readily be grown on methane in batch and continuous fermenter culture, achieving high cell densities (e.g., 1–5 g dry weight per liter) with specific growth rates approaching 0.2 h^{-1}. There are now reports on other fast-growing traits, such as *Methylomonas* and *Methylomicrobium (currently Methylotuvimicrobium)*, which display growth rates around 0.4 h^{-1} (summarized in Kalyuzhnaya 2016).

Patience is often required in the enrichment and isolation of methanotrophs, and while some methanotrophs yield colonies within 5–7 days, many are slow-growing. For example, it can take 2–4 weeks before enrichments become turbid or colonies appear on agar plates during the isolation of more unusual methanotrophs from exotic environments such as hot springs (e.g., *Methylocaldum*, *Methylocella*). The use of dilute mineral salts medium has been successful in the isolation of moderately acidophilic methanotrophs such as *Methylocapsa* and *Methylocella* from peat bogs (Dedysh 2002). Clearly, the choice of medium and enrichment conditions are important for the isolation of new methanotrophs, and certainly the original NMS medium which has been widely used over the last 40 years can bias selective enrichment of methanotrophs. For example, *Methylococcus capsulatus* (Bath) can easily be re-isolated from the Roman Baths in Bath, UK and forms the dominant methanotroph in enrichment cultures with NMS. However, molecular ecological studies have shown that this is not the dominant methanotroph in this environment, and alternative media and approaches need to be employed to isolate other methanotrophs present in this moderately thermophilic environment (Murrell et al., unpublished). Many methanotrophs do not readily store at −80 °C or in cryoprotectants such as glycerol, and so subculturing on agar plates every few weeks is recommended to maintain healthy, viable cultures.

7 Genomics and Genetics

Methylococcus capsulatus (Bath) is probably the best characterized methanotrophs and has been the "workhorse" organism for researchers studying the biology of methane oxidation for over 40 years. The genome of *Mc. capsulatus* (Bath) has been sequenced (Ward et al. 2004) and thus provides a metabolic "blueprint" which enables further study of one-carbon metabolism and the regulation of methane oxidation in this bacterium. Several genomes of different methanotrophic cultures are now available, providing the necessary background information to carry out comparative genomics and to determine experimentally the molecular basis for obligate methanotrophy (reviewed in Wood et al. 2004) in many methane-oxidizing bacteria.

Molecular genetic systems for obligate methanotrophs have been slow in development, largely due to their obligate nature (only growth on methane makes it difficult to select for alternative phenotypes). Fortunately there are now a number of genetic tools available to carry out mutagenesis and expression studies with *Mc. capsulatus* (Bath) and other methanotrophs (e.g., see Csaki et al. 2001, 2003; Sharpe et al. 2007; Theisen et al. 2005). These techniques allow the introduction of broad-host-range plasmids carrying homologous and heterologous genes into methanotrophs, promoter probe fusions, transposon mutagenesis, and mutagenesis by marker-exchange. Homologous expression systems are also available for certain methanotrophs (e.g., see Borodina et al. 2007) in order to study structure and function of MMO. Whole-genome metabolic models coupled with flux-balance modeling for three methanotrophic species have recently been reported. The models summarize the current knowledge of methane utilization, enabling in silico simulation of the catalytic capabilities of methanotrophic cells and streamlining metabolic engineering (reviewed in Akberdin et al. 2018a and in Henard and Guarnieri 2018). The genomics and post-genomics of methanotrophs have been described in Akberdin et al. 2018a, b and in Chistoserdova and Kalyuzhnaya 2018.

8 Research Needs

1. *Core biochemistry and molecular regulation.* While key steps of microbial methane utilization are now well established, there are knowledge gaps which hold back further developments in industrial applications. They include the limited knowledge of the initial steps of methane oxidation driven by pMMO, the lack of molecular data on stress responses, and the impact of environmental perturbations, specifically O_2-limitation. While methanotrophy is highly metabolically redundant, and a number of minerals, such as iron, copper, rare earth

elements, and tungsten, have been shown to impact the carbon flow across primary metabolic pathways, the detailed vision of interactions and responses to variability of nutrients remains to be established. Membrane biogenesis is also a much neglected area of research, and for most methanotrophs, the exact function of these unusual intracytoplasmic membranes and how their synthesis is regulated represents an interesting challenge which will be aided by genome sequence information. The mechanisms of intracellular oxygen production in some anaerobic methanotrophs also await further validation and might provide a great tool for methane-based fermentation processes. The nature of as-yet uncultivated methanotrophs which grow on atmospheric concentrations of methane also needs to be addressed. The extremophile methanotrophs which can grow at very high or very low pH values provide a means to study mechanisms by which these methanotrophs survive and even thrive at extremes. The differences between the metabolism of mesophilic and thermophilic methanotrophs can now also be studied in detail using emerging genome information as metabolic blueprints. Thermophilic methanotrophs may represent a new source of enzymes with potential in biocatalysis.
2. *Taxonomical and functional diversity.* Molecular ecology studies have revealed that there are considerably more methanotrophs present in the environment than are represented in culture collections. Considerable effort now needs to be made in the enrichment and isolation of novel methanotrophs, particularly those from more extreme environments. There are few marine methanotrophs in culture, and their role in the marine environment warrants further study.
3. *Quorum sensing and community interactions.* The mechanisms of cell-to-cell communication in methanotrophic communities are an emerging area which could provide new insights into ecosystem-level methane cycling. A number of methanotroph-associated viruses have been discovered, but their role in the regulation methanotrophic community structures and methane fluxes remains elusive. The symbiotic methanotrophs, which have mostly been characterized in marine environments, have no representatives in culture yet and thus will provide a fascinating area for future study, particularly with respect to the involvement of the methanotroph in supplying nutrients to the host and vice versa. Thus far, there is only one example of a methanotroph-plant symbiont, and again such symbioses will provide a fruitful area for study.

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